[AIDTM]

POST GRADUATE DIPLOMA IN MANAGEMENT (PGDM)

INDIVIDUAL ACADEMIC REPORT

AI IN SUSTAINABILITY & GREEN INITIATIVES

Course: Introduction to Artificial Intelligence

STUDENT DETAILS

Name	Dhaval Narendra Sharma
Roll Number	20262001
Programme / Trimester	PGDM (AIDS) / 1
Course Faculty	Chintan Patel
Academic Year	2026–2028
Date of Submission	18th October 2026

Executive Summary and Introduction

Executive summary

Large organisations operate facilities, equipment, transport networks and supply chains that consume energy, water and materials. Their environmental impact is shaped by thousands of daily decisions, yet information is often spread across meters, maintenance systems, spreadsheets, procurement records and periodic sustainability reports. This makes it difficult to identify emerging problems early or understand the operational causes behind a change in performance. This report proposes an AI-driven Sustainability Intelligence System for a hypothetical large Indian organisation. The system combines operational data, machine-learning models, search and optimisation methods, knowledge representation, rule-based reasoning and expert-system recommendations to support practical environmental decisions.

Purpose and scope

The proposed framework focuses on three connected priorities: energy optimisation, waste reduction and environmental monitoring. It is designed to help managers forecast resource demand, detect unusual consumption, identify likely causes, prioritise interventions and measure whether those interventions work. It is not presented as an existing system implemented by a named company. Public sustainability reports are used only to establish the business relevance of the problem, while the proposed design is an academic application of the AI concepts covered in the Introduction to Artificial Intelligence course.

Why sustainability requires better decisions

A monthly electricity bill may show that consumption increased, but it may not explain whether the reason was higher production, hotter weather, inefficient equipment, a faulty sensor or an operating schedule that was not followed. Waste totals have a similar limitation: a high figure may reflect production volume, defective material, poor planning, damaged storage or weak segregation. A useful system must connect measurements with context and provide a clear next step, rather than simply displaying more numbers.

AI as a decision-support capability

AI supports sustainability by analysing images, predicting consumption, classifying materials, detecting unusual patterns, and optimising routes and schedules. Rule-based and expert systems apply policies and domain knowledge to explain recommendations, while intelligent agents monitor data and automate approved routine actions.

Management perspective

AI should support, not replace, human decision-making in sustainability. Managers must verify unusual readings and assess safety, production and compliance risks before acting. By making evidence, uncertainty and constraints transparent, the system ensures accountable decisions and measurable outcomes.

Report structure

The following sections define the business problem and data foundation, explain the AI architecture, and apply core course concepts to energy optimisation, waste reduction and environmental monitoring. The discussion includes state-space search, breadth-first search (BFS), depth-first search (DFS), heuristic search, hill climbing, genetic algorithms, regression, classification, clustering, anomaly detection, computer vision, knowledge graphs, rule-based systems, expert systems and intelligent agents. The final sections consider implementation, model evaluation, governance and limitations. Supplementary diagrams and KPI tables are placed in the appendix.

1. Problem Definition, Industry Context and Data Foundation

Problem definition

A diversified organisation may run factories, warehouses, offices, logistics operations, utilities and renewable-energy assets. Electricity demand varies with production schedules, weather, equipment condition and operating hours. Water consumption can rise because of leakage or a process deviation. Waste may increase due to poor material planning, product defects, packaging choices or inadequate segregation. Transport emissions depend on routes, vehicle utilisation, load, fuel type and congestion. These activities are interdependent, so a decision that improves one indicator can affect another. The organisation needs a common way to interpret these relationships and prioritise action.

Limitations of traditional monitoring

Traditional sustainability processes often rely on periodic audits, fixed thresholds and manual spreadsheet consolidation. These methods can establish accountability, but they are mainly retrospective. A monthly report can reveal an increase after the opportunity to prevent it has passed. Fixed thresholds may miss a developing problem that remains below a limit but is abnormal for a particular machine or shift. Disconnected systems also create blind spots: the energy team may see electricity readings while maintenance records show an overdue service, and production data explains why output changed. Without integration, managers may draw incomplete conclusions.

Objectives of the proposed system

The system aims to establish reliable data, forecast resource needs, detect anomalies, reduce waste, monitor environmental indicators, and recommend practical actions. Success will be measured through outcomes such as lower energy use, reduced scrap, earlier leak detection, and improved corrective actions—not simply the number of AI models or alerts deployed.

Data sources and data types

Potential inputs include smart-meter readings, fuel and water records, temperature, pressure, flow and vibration sensors, equipment runtime, production quantity, maintenance history, waste audits, transport logs, supplier information, weather data and renewable-energy generation. Structured data may come from databases and spreadsheets; semi-structured records may come from system logs; unstructured information may include inspection notes, images and documents. Quantitative measurements support forecasting and regression, while qualitative notes can provide context after appropriate processing. Each record should include a timestamp, measurement unit, location or asset identifier and source information wherever possible.

Data preparation and quality

Data preparation is essential for reliable AI predictions. Missing values, duplicate records, inconsistent units, incorrect timestamps, and faulty sensors can lead to misleading results. For example, kilowatts measure power, while kilowatt-hours measure energy, so they must be compared appropriately. The system should standardise data, document changes, and flag uncertainties instead of silently correcting them. Unreliable data should never produce overly precise or misleading recommendations.

Industry context and benchmark use

Reports from Google, ITC Limited, Schneider Electric, and Reliance Industries highlight the importance of energy efficiency, resource management, and environmental measurement across different industries. Google focuses on data-centre energy efficiency, while ITC and Reliance disclose sustainability initiatives across their operations. Schneider Electric emphasises energy management solutions. These examples serve as industry benchmarks, not proof that the exact proposed system is already

implemented by these companies. The key takeaway is that AI creates real sustainability value when its insights support practical actions, measurable targets, and accountable decisions.

2. Proposed AI Architecture and Intelligent Problem-Solving

A layered decision-support system

The proposed system follows a connected process, from data collection to outcome measurement. Sensors and business systems provide data, which is checked and standardised before AI models forecast demand or detect unusual patterns. Knowledge graphs connect these findings to equipment, processes, and sustainability targets, while rules check them against policies and thresholds. An expert system then recommends suitable actions for managers to approve or adjust. The final results are recorded to evaluate whether the actions improved sustainability performance.

Problem formulation and state-space search

Search begins by defining a problem in terms of an initial state, possible actions, a transition model, a goal state and a path cost. In sustainability, a state might describe the operating schedule of a set of machines, the current route plan for waste collection or the allocation of flexible energy demand. Actions represent possible changes, such as moving a process to another time slot or assigning a collection vehicle to a different route. A goal could be to meet production demand while lowering cost or emissions. Constraints define what cannot be violated, including safety limits, equipment availability and delivery deadlines. Clear problem formulation prevents the algorithm from optimising the wrong objective.

Breadth-first search and depth-first search

Breadth-first search (BFS) explores all options at one level before moving deeper. When every action has the same cost, it can find the solution with the fewest steps, but it may require more memory. Depth-first search (DFS) follows one path until it reaches a limit, then backtracks. It uses less memory but may not find the shortest solution. In sustainability, both methods can explore machine schedules, waste collection routes, and operational choices. However, larger problems often require more specialised optimisation techniques.

Heuristic search and A*

Heuristics help search algorithms choose the most promising path towards a goal. Greedy best-first search uses an estimate of how close a state is to the goal, while A* considers both the cost already incurred and the estimated remaining cost. For example, these methods can help plan waste-collection routes or energy-efficient machine schedules. However, the estimates must be carefully designed, and the objective should balance cost, emissions, safety, and operational needs rather than focusing on cost alone.

Optimisation methods

Search algorithms explore possible paths, while optimisation methods compare solutions to find better outcomes. Hill climbing improves a solution step by step but may get stuck at a local optimum. Simulated annealing can accept temporary setbacks to explore better options, while tabu search avoids repeatedly visiting recent solutions. Genetic algorithms use selection, crossover, and mutation to develop new solutions. These methods can improve machine schedules, energy allocation, and transport routes, but their results must be tested against real constraints and compared with a baseline before implementation.

Choosing methods responsibly

No single search algorithm works best for every sustainability problem. BFS and DFS are useful for exploring simpler problems, while A* works well when a reliable heuristic is available. Hill climbing and genetic algorithms can handle complex problems but may not always find the best possible solution.

The choice depends on problem size, available time, data quality, and the cost of errors. A practical approach is to start with a simple method and adopt more advanced techniques only when they show clear improvements.

3. Machine Learning, Knowledge Graphs and Reasoning Systems

Regression and time-series forecasting

Regression helps predict values such as energy use, water demand, fuel consumption, and waste based on factors like production levels, weather, and equipment condition. Time-series forecasting uses past data to predict future demand and identify recurring patterns, helping organisations plan resources and manage peak usage. Models should be tested against simple benchmarks, such as recent averages, to ensure they provide meaningful improvements.

Classification and decision trees

Classification groups data into categories, such as normal or abnormal operations, recyclable or non-recyclable waste, and different levels of maintenance risk. Decision trees make predictions by following a series of questions. For example, they can flag a machine for inspection based on its energy use and vibration levels. Since their decisions are relatively easy to understand, they are useful for operational teams. However, they can overfit data, so regular testing and review are important.

Clustering and anomaly detection

Clustering groups similar data, helping organisations compare facilities based on energy use, operating hours, and production patterns. It can highlight unusual groups or identify where an audit may be needed. Anomaly detection flags unusual behaviour, such as a sudden rise in overnight electricity use. However, these findings are only signals for further investigation, not definite proof of waste or equipment failure.

Knowledge graphs and ontologies

A knowledge graph connects important information, such as facilities, machines, energy meters, maintenance records, and sustainability targets. This helps AI link a prediction to its operational context. For example, a rise in a machine's energy use can be checked against its maintenance history and production line. An ontology defines these entities and relationships consistently, making data easier to understand, compare, and trace across different sites.

Rule-based AI systems

A rule-based system applies predefined logic, often written as IF–THEN statements. For example: IF water flow remains high while a production line is stopped, THEN raise a possible-leak alert. IF the verified emissions indicator exceeds an approved threshold, THEN notify the responsible environmental manager. IF a waste-sorting model reports low confidence, THEN send the item for human inspection. Such rules are transparent and do not require training data, making them useful for stable policies and clear thresholds. However, they do not learn automatically and can become brittle when rules are incomplete or operating conditions change. Rules should be documented, tested and maintained by accountable owners.

Expert systems and inference

An expert system uses specialised knowledge and rules to support decisions like a human expert. It includes a knowledge base, an inference engine, and an explanation facility that shows why a recommendation was made. In sustainability, it can combine equipment knowledge, policies, and AI predictions to suggest suitable actions. Forward chaining starts with known facts to reach a conclusion, while backward chaining checks whether the conditions for a possible conclusion are met. Unlike a basic rule engine, an expert system also uses context to explain its recommendations.

Combining learning and reasoning

Learning-based AI identifies patterns in data, while rule-based systems apply predefined conditions. Combining both helps produce more useful recommendations. For example, a model may detect unusual energy use in a compressor, while maintenance records and rules help identify a possible cause and suggest an inspection. The system should explain its reasoning, highlight uncertainty, and leave important decisions to human experts.

4. AI for Energy Optimisation and Predictive Maintenance

Energy as an operational decision

Energy optimisation is not simply about using less electricity. Energy demand depends on production levels, operating hours, weather, equipment condition, and renewable energy availability. Any reduction must maintain safety and production requirements. AI can estimate expected consumption, detect unusual usage, and suggest improvements by considering cost, emissions, and reliability. Measures such as kilowatt-hours per unit produced help managers compare energy efficiency across periods with different production levels.

Demand forecasting and regression

Historical meter readings, shift schedules, weather, production plans and calendar patterns can be used to forecast demand for the next hour, day or shift. Regression can estimate the relationship between consumption and explanatory variables, while time-series models can capture recurring patterns. These estimates help managers anticipate peak demand, schedule flexible processes and plan the use of renewable generation. Forecasting performance should be evaluated on data not used for model training and compared with a simple baseline. If a model is consistently inaccurate during unusual weather or production changes, the system should flag the limitation rather than present the forecast as certain.

Anomaly detection in equipment

Equipment may gradually become less efficient because of wear, poor calibration, leakage, blocked filters or maintenance issues. Anomaly detection can compare current energy, temperature, pressure, vibration and runtime with expected patterns for the asset and its operating context. For example, a pump consuming more power while moving a similar volume of water may warrant inspection. A compressor that operates for unusually long periods after production ends may indicate air leakage or an incorrect control setting. The alert should include the observed deviation, comparison baseline, relevant operating conditions and recommended checks. It should not declare a fault without validation.

Predictive maintenance

Predictive maintenance uses equipment data and past maintenance records to identify possible failures before they occur. AI models can classify equipment risk or estimate remaining operating time when sufficient data is available. This helps reduce downtime, safety risks, and energy waste. However, when historical failure data is limited, simple monitoring rules may be more reliable than complex models. Alerts should also be assigned, resolved, and reviewed through the organisation’s maintenance process.

Scheduling and optimisation

Optimisation algorithms compare equipment schedules using factors such as energy demand, electricity prices, renewable energy availability, and production deadlines. For example, they can shift flexible processes to periods with more solar power or reduce peak demand by staggering equipment start times. Methods like hill climbing and genetic algorithms can help find better schedules. However, every recommendation must meet safety, maintenance, production, and quality requirements to ensure it is practical to implement.

Buildings, utilities and renewable energy

AI can improve building operations by using temperature, humidity, occupancy, and weather data to manage cooling and reduce unnecessary energy use. It can also detect unusual behaviour in water pumps and compressed-air systems. Renewable-energy forecasts can help schedule flexible activities when more clean energy is available. These changes must maintain comfort and equipment safety. Testing the system in one facility first can help identify problems before expanding it.

Measuring real impact

Energy savings should be measured against a baseline that accounts for changes in production, weather, and operating hours. Organisations must distinguish actual savings from model estimates and avoid assuming every improvement comes from AI. Key measures include energy consumption, energy intensity, peak demand, forecast accuracy, equipment downtime, renewable-energy use, and verified cost savings. Reviewing false alerts and maintenance results helps managers understand what worked and whether the approach needs improvement.

5. AI for Waste Reduction, Circularity and Environmental Monitoring

Understanding waste sources

Waste often results from wider process issues, such as over-ordering, poor forecasting, storage problems, product defects, or contamination. AI can help identify where waste occurs and which conditions contribute to it. Organisations should record waste consistently by material, source, quantity, and destination. This helps distinguish waste prevention from recycling and supports fair performance comparisons across facilities.

Computer vision and convolutional neural networks

Computer vision can analyse images of waste, material defects, and sorting stations. Convolutional neural networks (CNNs) learn visual patterns to classify materials and identify visible defects. For example, they can help separate waste into categories or flag contaminated materials for review. Accuracy depends on image quality, lighting, camera position, and training data. Unclear cases should be checked by trained staff, and image collection must follow privacy and workplace policies.

Forecasting and waste prevention

Demand forecasting helps organisations estimate future material needs, reducing excess inventory and expired stock. Regression can predict material use and waste, while classification can identify batches at higher risk of defects. Clustering helps find production lines with similar waste patterns. These insights guide audits and improvement efforts, but they do not automatically explain the cause. Further investigation is needed to determine whether waste results from machine settings, material quality, workload, or other factors.

Circular economy applications

A circular economy seeks to preserve the value of materials through reduction, reuse, repair, remanufacturing and recycling. AI can help identify reusable components, forecast the availability of recovered materials, match waste streams with possible users and improve reverse-logistics planning. Search and route optimisation may support collection planning, while knowledge graphs can connect material sources, quality specifications, facilities and potential reuse pathways. A digital match does not guarantee that reuse is safe or feasible. Material quality, contamination, transport cost, legal requirements and lifecycle impacts must be checked before a proposed reuse route is approved.

Environmental monitoring

Environmental monitoring may include electricity, fuel, water, wastewater, air quality, greenhouse-gas emissions and other indicators relevant to the organisation. Sensors can provide frequent readings, while periodic audits or laboratory tests may provide validated measurements. AI can combine these sources to identify unusual patterns, estimate whether a threshold may be crossed and prioritise investigation. The system should retain the original data, measurement units, timestamps and calculation methods. Sensor failure must be considered alongside a genuine environmental event, because an incorrect reading can create a false alarm or conceal a real issue.

Rule-based escalation and expert review

A rule can state that a sustained rise in water flow outside operating hours requires a leak investigation, or that a verified environmental reading beyond an approved limit must be escalated. An expert system can then combine the alert with asset information, historical readings, weather, maintenance records and known operating changes. The resulting recommendation should identify the evidence, uncertainty, responsible person and required response. High-risk environmental incidents must follow established emergency and compliance procedures; an AI system should not delay escalation while attempting to reach its own conclusion.

Environmental performance and reporting

Key indicators include water use, waste per unit of output, recycling contamination, greenhouse-gas emissions, energy intensity, and response to alerts. Each indicator should have a clear definition, unit, data owner, calculation method, and reporting frequency. Estimated emissions must be distinguished from verified figures, with assumptions properly documented. Dashboards help track trends, but reliable data, clear accountability, and traceable evidence are essential for accurate reporting.

6. Intelligent Agents, Implementation and Responsible AI

Intelligent agents in sustainability

An intelligent agent monitors its environment, analyses information, and takes actions to achieve a goal. In sustainability, it can track energy readings, flag unusual patterns, recommend equipment schedules, and prepare environmental reports. Repetitive tasks, such as data-quality checks, can be automated, but actions affecting safety, production, or regulatory reporting should require approval from an authorised person.

Planning and multi-step decisions

Planning involves choosing the right sequence of actions to achieve a goal while meeting necessary constraints. In sustainability, this could mean detecting an energy issue, checking equipment records, arranging an inspection, scheduling repairs, and measuring the results. Each step must follow the correct order and receive the required approvals. The plan should also account for real-world limitations and include alternatives if an action fails or circumstances change.

Implementation roadmap

A phased approach is more practical than implementing AI across all facilities at once. First, identify sustainability priorities, available data, and responsible decision-makers. Next, select a pilot project with measurable benefits, prepare the data, and test the model against simpler methods. Then, integrate the results into existing workflows, train users, and measure performance. Expand the system only when the pilot demonstrates reliable results, manageable risks, and clear operational value.

People, process and adoption

AI implementation also requires proper change management. Facility teams should understand why alerts are raised and what actions are needed, while sustainability and IT teams ensure consistent data and secure systems. Users should be able to report false alerts, reject unsuitable recommendations, and explain changes to decisions. Their feedback should be reviewed before improving the system. For successful adoption, AI must reduce workload and provide clear benefits that fit existing operational needs.

Responsible AI and explainability

Responsible AI should focus on accuracy, transparency, privacy, security, fairness, and accountability. Industrial data may contain sensitive information about employees, production schedules, or business operations, so access should be restricted and data collection kept necessary. AI recommendations should explain their reasoning, indicate uncertainty, and flag missing or unreliable data. Important decisions must remain open to human review and follow established approval procedures.

7. Evaluation, Recommendations and Conclusion

Evaluation framework

The system should be evaluated across four areas: environmental, operational, technical, and governance. Environmental indicators include energy and water use, waste reduction, recycling quality, and verified emission savings. Operational measures cover downtime, production targets, response time, and corrective actions. Technical evaluation focuses on model accuracy, data quality, and reliable alerts, while governance checks traceability, human oversight, and regular model reviews. Defining these measures before the pilot begins ensures that results are assessed fairly and consistently.

Recommendations

Start with a clear sustainability problem and improve data quality before building AI models. Use simple, explainable methods where possible, and apply search or optimisation only when objectives and constraints are clearly defined. Combine predictions with operational knowledge to make recommendations more useful. Maintain human oversight and proper records for important decisions. Expand the system only when a pilot demonstrates measurable improvements against a reliable baseline.

Limitations

The system's effectiveness depends on reliable data, suitable AI methods, and user adoption. Models may not work equally well across facilities because equipment, climate, and operating practices differ. Unusual readings may result from faulty sensors rather than real problems, and correlations do not always explain the cause. Similarly, optimisation may suggest impractical solutions if constraints are missing. Regular testing, human review, and transparent reporting are essential to ensure reliable results and demonstrate genuine sustainability benefits.

Conclusion

AI can help organisations make faster, data-driven sustainability decisions instead of relying only on past reports. Forecasting improves resource planning, anomaly detection identifies unusual patterns, computer vision supports waste sorting, and optimisation helps balance cost, emissions, and production needs. Search algorithms explore possible actions, while knowledge graphs and expert systems connect predictions with operational knowledge. Intelligent agents support continuous monitoring while keeping humans accountable. Ultimately, success depends on measurable improvements in resource efficiency, waste reduction, environmental monitoring, and decision-making—not on technical complexity alone.

Appendix A. Proposed Architecture and Knowledge Relationships

System flow: Operational and environmental data → data validation and integration → machine-learning and search methods → knowledge graph → rule-based checks → expert-system recommendation → human review and action → outcome measurement.

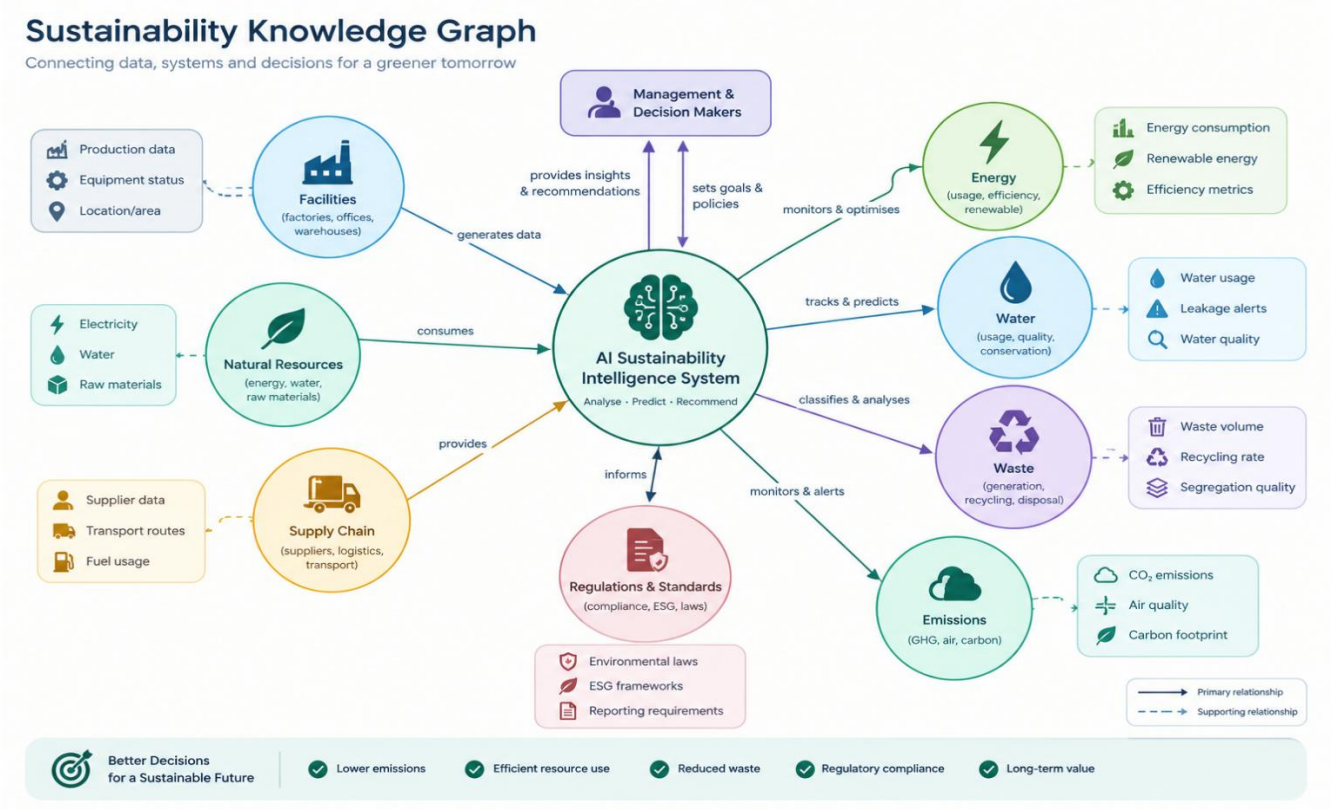
Example knowledge relationships: Facility contains Equipment; Equipment is measured by Meter; Meter produces Consumption Reading; Equipment has Operating Schedule; Equipment is associated with Maintenance Record; Production Line creates Output and Waste; Waste Stream has Material Type and Destination; Environmental Indicator is calculated from validated readings; Recommendation is reviewed by Responsible Manager; Corrective Action is evaluated using Post-Action Measurement.

Example decision path: Receive and validate a reading; compare it with a context-adjusted baseline; check sensor quality, schedule and known events; apply relevant rules and operational constraints; generate an explanation and confidence estimate; route consequential recommendations to an authorised person; record the action and evaluate subsequent measurements.

Search and optimisation example: A facility must schedule flexible processes while meeting output deadlines. The initial state is the current schedule; actions move processes between approved time slots; the goal is to meet output requirements while reducing energy cost and emissions. Constraints include machine availability, staff coverage, safety and delivery deadlines. BFS/DFS illustrate path exploration for a small state space, while heuristic search or optimisation is more suitable when combinations become large.

Expert-system example: IF electricity use is unusually high for a sustained period AND the equipment is operating in the relevant period, THEN review the asset's maintenance history. IF an overdue task is found AND the deviation is confirmed, THEN recommend inspection and assign the alert to the asset owner. IF sensor quality is poor, THEN validate the data before recommending a physical intervention. The explanation facility records the facts and rules supporting the recommendation.

Appendix C. Knowledge Graph for Sustainability Intelligence



Appendix B. Suggested KPIs and Model Review Checklist

Area	Indicator	Purpose
Energy	Energy intensity	Energy used per unit of output, adjusted for operating context.
Energy	Forecast error	Checks whether predicted demand improves on a simple baseline.
Energy	Verified savings	Measured reduction against an adjusted baseline.
Waste	Waste intensity	Waste generated per unit of output or service.
Waste	Contamination rate	Share of material rejected from a recycling stream.
Environment	Data completeness	Share of expected readings available and passing checks.
Environment	Closure time	Time from validated alert to documented resolution.
Governance	Human review rate	Share of consequential recommendations reviewed by a person.

- Confirm the model has a defined purpose, owner and baseline.
- Check data completeness, units, timestamps, sensor reliability and data lineage.
- Compare performance with a simple baseline and test on unseen data.
- Review false positives, missed events and differences across sites.
- Document assumptions, confidence, limitations, rules and escalation paths.
- Check privacy, security, permissions and retention before integration.
- Reassess after changes to equipment, processes, climate or data patterns.
- Report measured outcomes separately from estimates and projections.

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